

Quantile Index Preservation under Bounded Score Perturbations

Research Architecture Runtime
operator/quantile

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Abstract

We study τ -quantile selection as the unique strict k -th order statistic on a finite real score vector. If every pairwise gap exceeds 2ε , the selected index is invariant under $\|\delta\|_\infty \leq \varepsilon$. Lean `LEAN_FULL` on Mathlib `ℝ` (`REAL_MATHLIB`).

1 Operator

Definition 1. For $s \in \mathbb{R}^n$ ($n \geq 2$), index i is the unique strict k -th smallest when exactly k scores are strictly below s_i and no other index shares s_i . A τ -quantile uses $k = \lfloor \tau(n-1) \rfloor$.

2 Theorems

Theorem 1 (Preservation). If i is a unique strict k -th smallest and all pairwise gaps exceed 2ε , then every $\|\delta\|_\infty \leq \varepsilon$ perturbation preserves that status.

Theorem 2 (Sharpness). If some rival $j \neq i$ has $|s_i - s_j| \leq 2\varepsilon$, an admissible δ destroys unique strict k -th-smallest status of i (tie adversary).

3 Proofs

Pairwise order vs i is preserved under the gap hypothesis; `countLT` congruence and uniqueness follow. Sharpness forces a value-tie between i and a close rival.

4 Comparison

Relative to Argmax margin: same 2ε gap cost, but ranking/count geometry rather than unique-max. Relative to Threshold: multi-cut ordering instead of a single cut.

5 Lean / certificate

`Research.Operators.Quantile.Preservation` reducing to `OrderStat.KthMargin`; certificate `lean/certificate` (`LEAN_FULL`; Mathlib `ℝ`).

6 Limitations

Mathlib `ℝ` (`REAL_MATHLIB`); unique-value hypothesis; no privacy claims.